LS-CAT Management Board Meeting

Keith Brister



Agenda

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        Start Time: 1 PM CDT (2 PM EDT) May 5, 2015
             Audio: 1-888-850-4523
          Passcode: 190580
           Video 1: http://conf1.ls-cat.org
           Video 2: http://conf2.ls-cat.org
Physical Location: 9700 S. Cass Ave Bldg 436A, Argonne, IL
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Welcome John Tesmer as LS-CAT Management Board Chair!

1. Approval of Minutes
2. SAC Review Report
3. Status
4. Renewal
5. FY 2016 Budget
6. New Business

APS SAC Review of LS-CAT November 4, 2014

8:00	8:30	Executive Session (APCF Conference Room)	
8:30	8:45	Life Sciences Capabilities at the APS	Robert Fischetti
8:45	8:55	(Short walk to 436C)	
8:55	10:00	LS-CAT Science Overview, History, and Organization LS-CAT Design and User Program	Alfonso Mondragón Keith Brister
10:00	10:45	Biological methane oxidation	Amy Rosenzweig
10:45	11:00	Coffee Break	
11:00	11:30	LS-CAT Enabling Technologies	Keith Brister
11:30	12:30	The Future of LS-CAT: Discussion	Alfonso and Keith
12:30	1:30	Working Lunch with LS-CAT staff (without management)	
1:30	2:15	BNP Science	Gayle Woloschak
2:15	3:00	Structural mechanisms of DNA base excision and replication repair	Brandt Eichman
3:00	3:30	Coffee Break and Beamline Tour	
3:30	5:00	Executive Session	
5:00		Adjourn	

Committee Members

Britt Hedman, Chair, SSRL
Paul Adams, LBNL
Christopher Hill, University of Utah Medical School
Lisa Miller, BNL

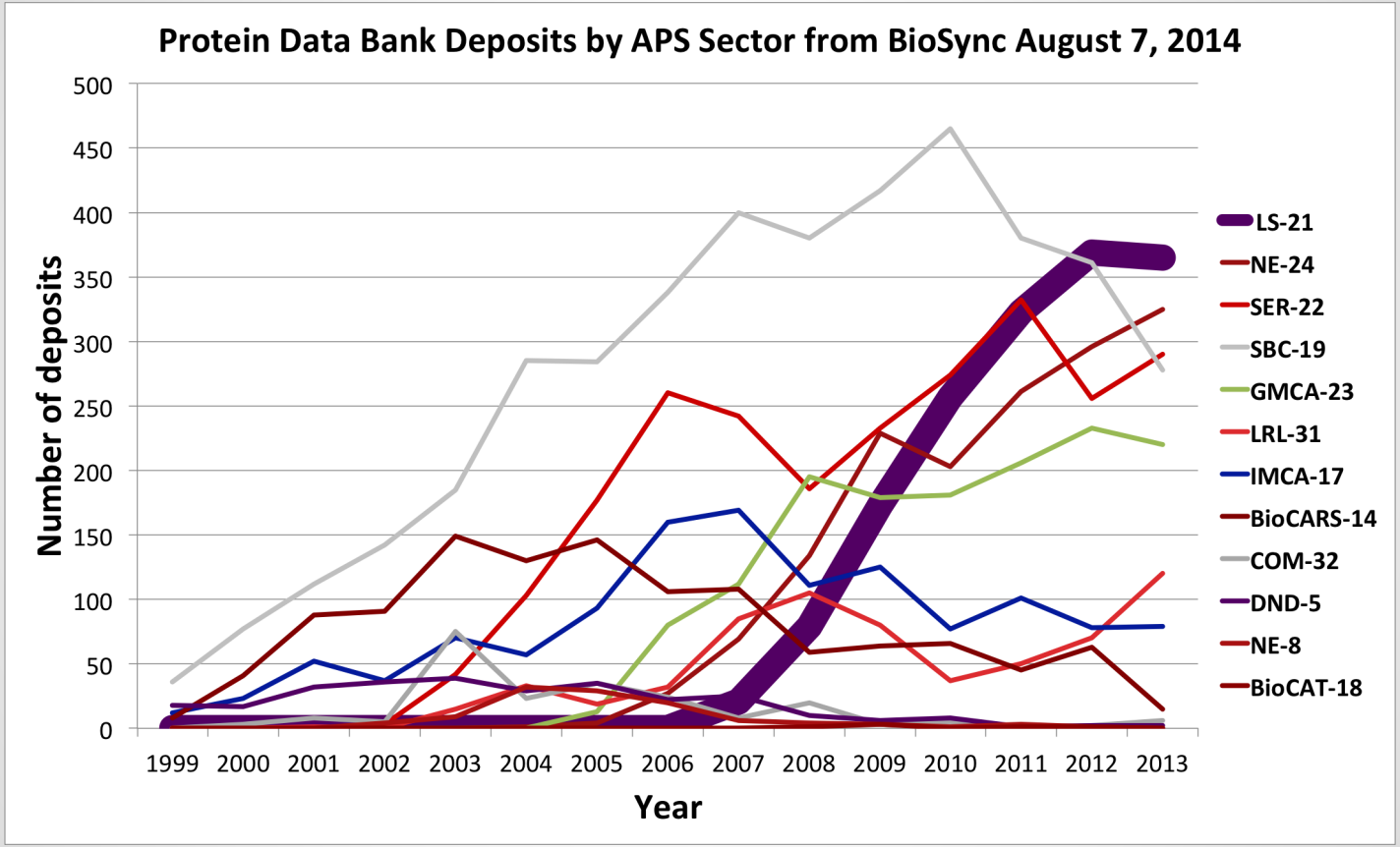
There were two parts of the of the review: a written summary and an oral presentation.

Evaluation Criteria

1. Scientific Mission and Impact
2. Beamline Productivity
3. Leveraging of specific APS characteristics
4. Effectiveness of beam time use
5. General User program quality
6. Adequacy and reliability of funding
7. Future Plan
8. Overall
9. Recommended Actions including recommendation to APS for continuation, a re-review before five years, or termination/decommissioning.

At the closeout the committee sounded impressed by all 7 of the specific areas above and will be recommending that we continue. We are hopeful that a written report will be available to us in January.

Beamline Productivity: PDB Deposits



Beamline Productivity: PDB Deposits (Continued)

The table on the right shows the PDB deposits from several of the most productive ID ports in the world integrated over a 4 year period.

This shows that LS-CAT is not only competitive with its APS peers but also with ID ports around the world.

When beamline productivity is measured by PDB deposits then LS-CAT is among the most productive or is the most productive when compared locally to other APS sectors or globally with other ID ports.

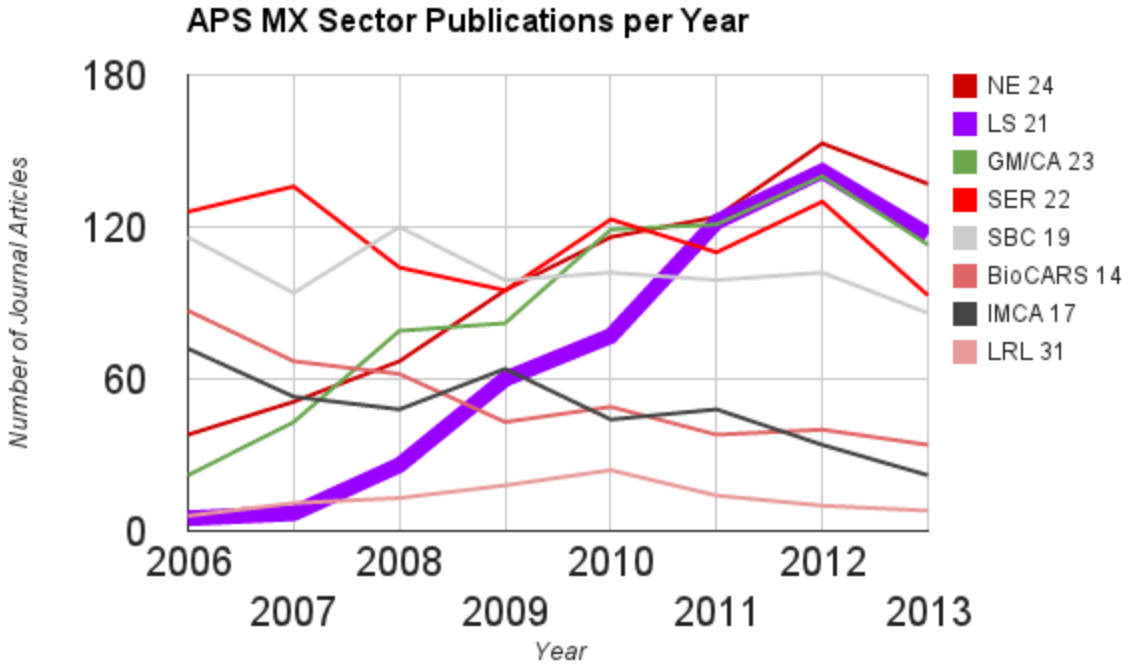
Insertion Device Port	PDB Deposits 2011 – 2014 (BioSync 8/7/14)
LS-CAT APS 21-ID	1190
SLS 6	1076
ESAF ID-14	1032
NE-CAT APS 24-ID	946
SSRF 17	924
SBC APS 19-ID	910

Beamline Productivity: Journal Publications

While journal publication rate does not show LS-CAT as being the “best” (we’ re currently second best), it does show that LS-CAT’s publication rate rose quickly after first light to join its APS peers.

As of October 27, 2014 there were 636 journal publications in the APS database attributed to LS-CAT. Of these 156 used General User time. That is, **25% of LS-CAT journal publications are from General Users.**

Note that the APS queries the PDB to find journal articles. As more 2013 PDB deposits are processed we expect the 2013 journal numbers to increase as well.



SAC Review Report

CAT/XSD Group Name: LS-CAT (Life Sciences Collaborative Access Team)

CAT Director/Group Leader:

Scientific Director: Prof. Alfonso Mondragón
Manager: Keith Brister

Provide a numerical rating (1= exceptional performance, 2 = expected level of performance; 3 = weak performance, improvement needed) and brief comments for each of the following:

1. Scientific Mission and Impact – 1.5
2. Beamline Productivity – 1.5
3. Leveraging of specific APS characteristics – 1.5
4. Effectiveness of beam time use – 2
5. General User program quality – 1.5
6. Adequacy and reliability of funding – 1
7. Future Plan – 1.5
8. Overall – 1.5
9. Recommended Actions including recommendation to APS for continuation, a re-review before five years, or termination/decommissioning. (no numerical score given)

Specific comments are given in the sections below.

As presented in the detailed report below, the committee strongly recommends continuation of the LS-CAT sector.

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"Above average. Reasonably good or perhaps a bit better than that." - APS User Office

Specific comments are given in the sections below.

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Quotes from the report

“The LS-CAT team is to be congratulated for developing an ID-only sector where the beam lines are heavily used, in contrast to other sectors where bend magnet beam lines have typically received little use.”

“Presentations given at the review reinforced the impression that work being performed at LS-CAT is of central importance to very strong research programs.”

“At the review, the staff showed great enthusiasm and responsibility for scientific user support, keeping the instrumentation at high performance, and training. They carry a substantial workload and professed a high level of satisfaction with the organization and management. The staff and management are to be congratulated for providing such a smoothly and efficiently running operation, with very satisfied users and high scientific productivity.”

SAC Review Conclusions

The review panel concludes that the LS-CAT is an exceptionally highly productive CAT that provides excellent user support at well-performing beamlines. The users are very satisfied and productive, and the staff members display a positive can-do attitude. The LS-CAT members represent very strong scientific programs. LS-CAT has a dedicated educational/training component with a high number of Ph.D. theses as well as training for undergraduates, and intense user training workshops. The CAT focuses mainly on operations. R&D is sometimes challenging due to limitations in funding – but is well executed through collaborations as appropriate.

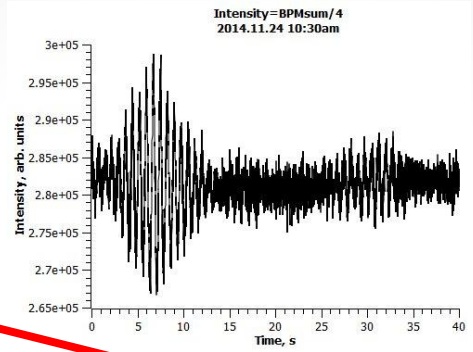
LS-CAT is well managed, and has a strong scientific and technical leadership. Funding is currently stable and adequate. The nano-probe is a unique instrument in the U.S. at present. It currently uses 25% time on 21-ID-D and will likely merit additional beam time in the future. This anticipated increase in demand should to be strategically addressed at the time of the APS upgrade.

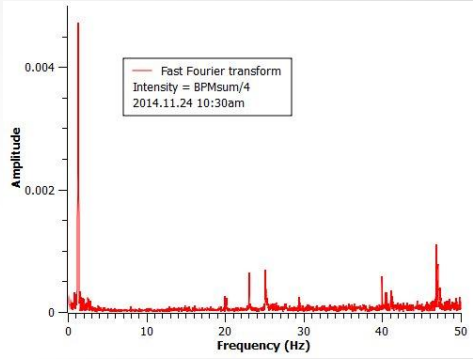
The review committee strongly recommends a continuation of the LS-CAT at sector 21-ID.

Status: 21-ID-D Mono

Last meeting I reported on an instability that affected operations on 21-ID-D. This was particularly a problem for the the Bionanoprobe.

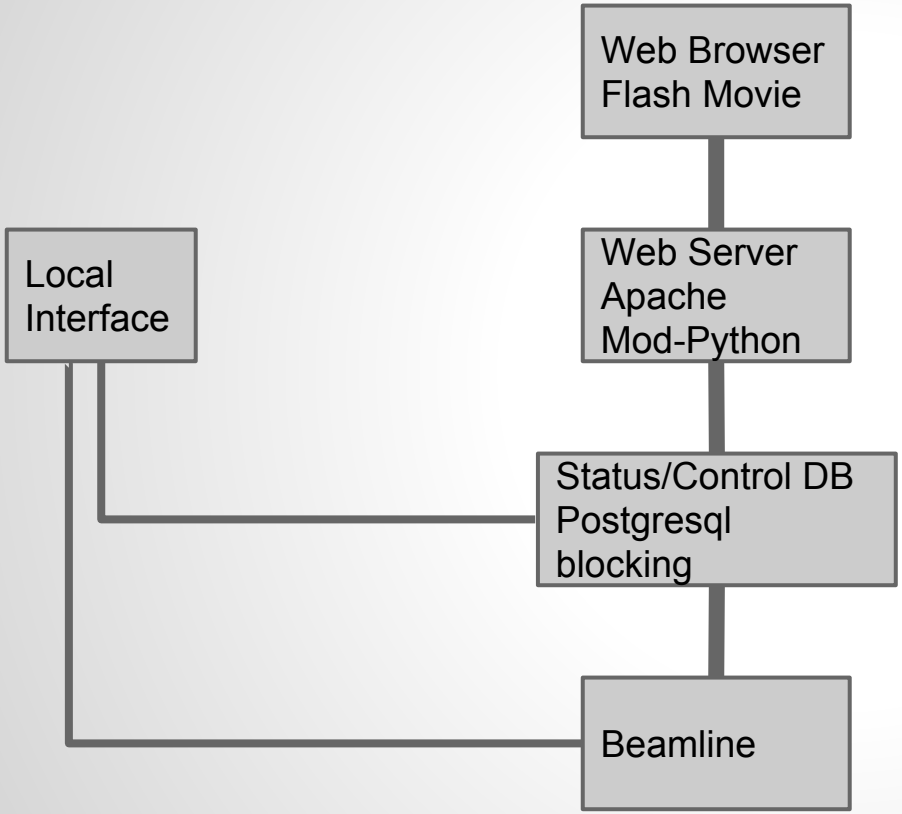
This problem has since been resolved: there was a small drop of water on the motor shaft that holds the LN2 impeller deep in the cryopump. Of course at LN2 temperatures this was a bit of very hard ice and enough to rip apart one of the bearings. This was the cause of the problem. The APS loaned us a cryopump while ours was being rebuilt. We'll be switching back to our cryopump this down period.



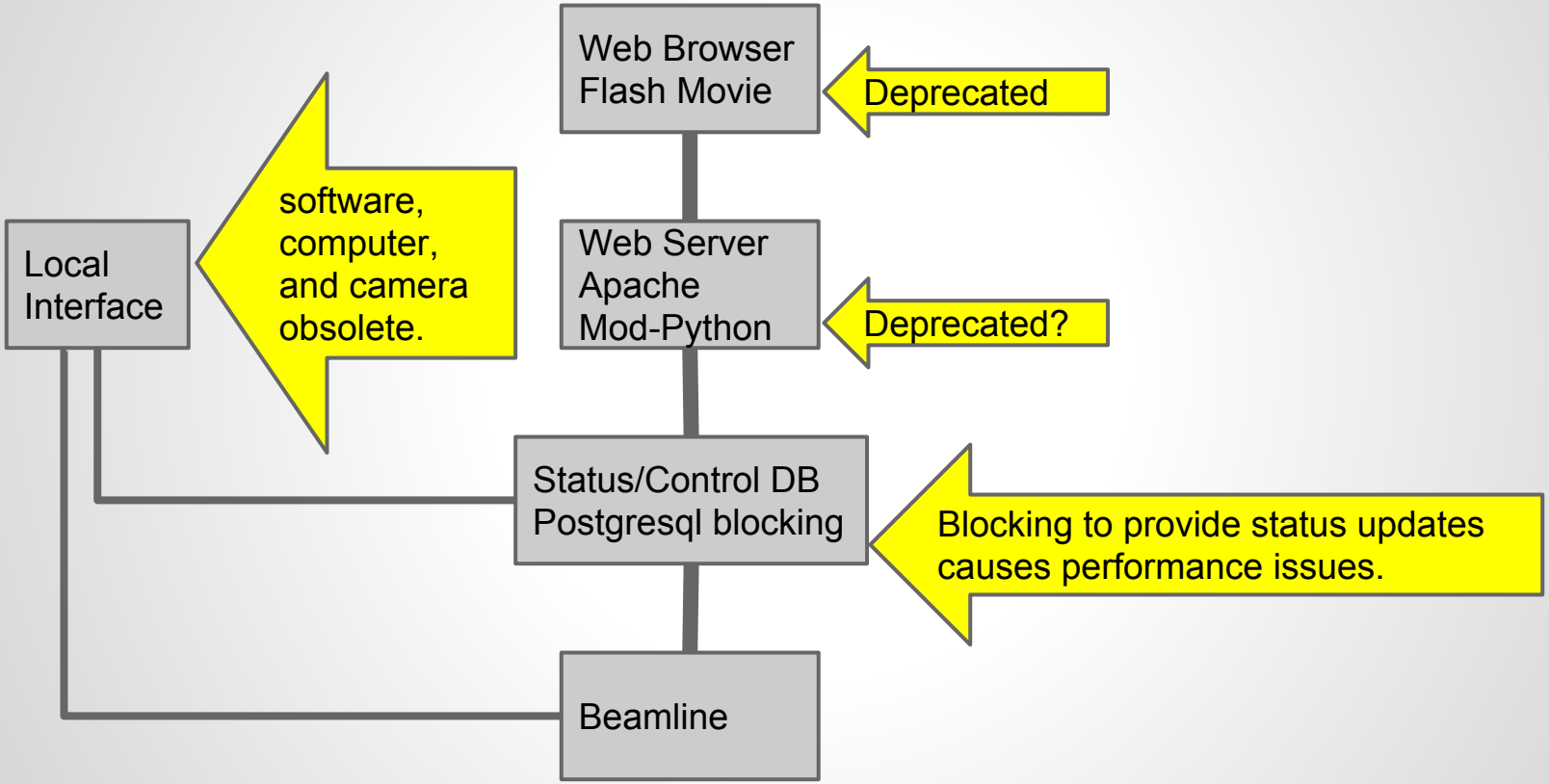




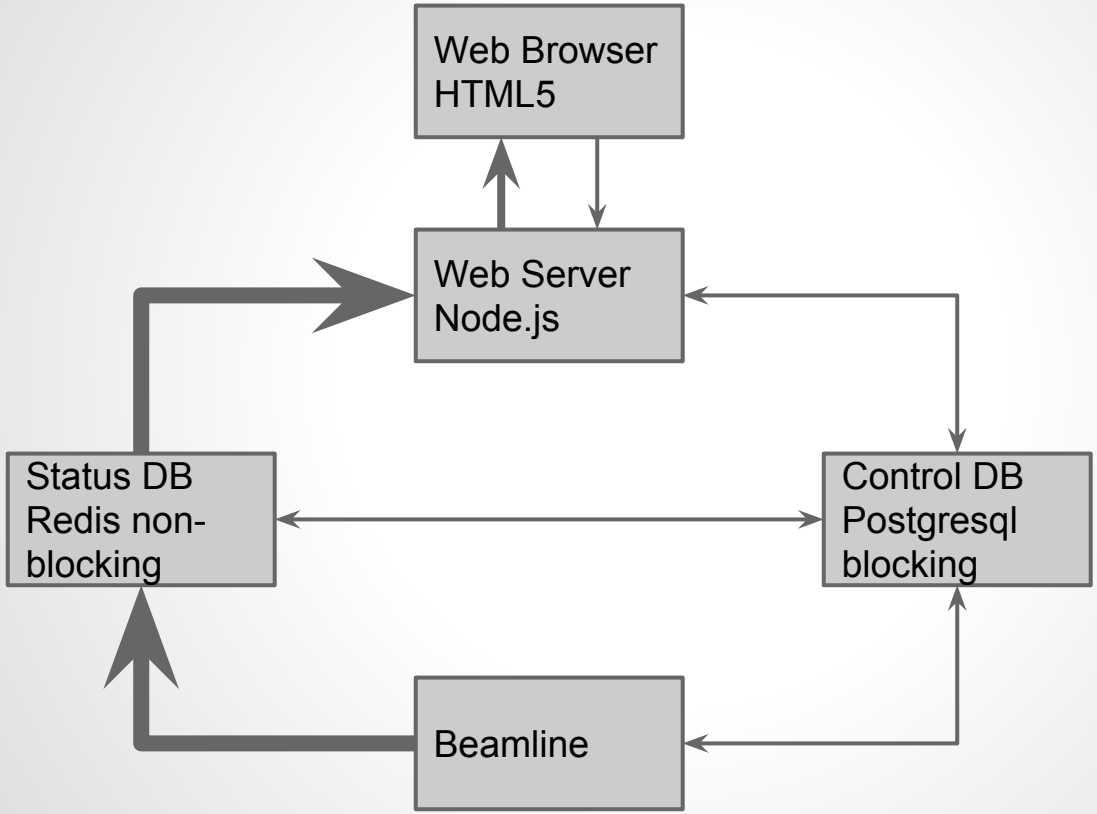
Old Control System Architecture



Old Control System Architecture



New Control System Architecture



New User Interface

This last run we started to roll out our new user interface. This was the most drastic change we have been made since 2006 when we introduced the MD-2 diffractometer on F and G.

What went right:

1. Although the first few days were pretty shaky we did not have to revert to the old software.
2. Most users like the new interface and are looking forward to the new features yet to come.
3. About 100 users participated in a crash course given over about a week at the beginning of the run.

New User Interface

What went as expected:

1. Sometimes users and/or staff would panic when they did not know how to do something that they used to be able to do.
2. Many bugs appeared only after users started to use the system.
3. System stability improved greatly after the first week or so of relative chaos.
4. Serious issues were discovered that could not be fixed quickly and easily, such as the evil “mounting of the wrong sample” bug.

New User Interface

What went wrong:

1. Between the software that users see and foundations of EPICS and database code there are about 2 dozen programs that keep the system running. One of these programs failed and caused communications problems between several key components of our data collection infrastructure. This was the root cause of many of the issues encountered this past run. This has now been fixed.

2. I had planned that centering would be done using short movies as in the previous remote access client. However, live centering (originally intended to only be used onsite) seems to work even better than remote centering for all but the slowest or most unstable of network connections. This is fortunate since the movie centering technique now only works (for reasons I can go into if you like) about half the time. The line segment data collection technique currently relies on movie centering and, hence, is not as easy to use as it should be.

Beamline software development cycle

1. (During run) Plan new fixes and features: whenever possible develop a “proof of principle” test to be sure that the chosen approach is sane.
2. (Start of Down Period) Carefully implement new fixes and features.
3. (2 weeks before next run) Finish up work. Take whatever steps are needed to bring project to a close including dropping new features.
4. (Last week before run) Do whatever it takes to make things usable, even if the end product falls short of what was desired.
5. (Beginning of run) Figure out how to work around all the new bugs that just appeared.
6. Go to step 1.

New User Interface

Planned fixes and features for 2015-2:

1. Line segment scans with live centering
2. Loop explorer (w/raster scans)
3. Improved, less confusing, layout
4. Documentation

As time permits:

1. Reintegration of diffraction images into the main user interface
2. Improved on-line analysis tools

Beamtime Allocations

For the time being I'm inclined to keep the beamtime allocation as though we have 6.5 shares instead of the current 5.5 shares. This leaves 8 days per station per run that we'll use to debug and advance the new user interface. This is particularly important as we gear up for acceptance of the Eiger detector due to arrive during the 2015-3 run.

LS-CAT Renewal

“Final” version of the amendment is ready: https://ls-cat.org/LS-CAT-FourthAmendment-1505.pdf

I would like the amendment to be fully signed and counter-signed by our November / December meeting. Note that it can take many months for NU to do anything so we do not have much time.

Key points:

1. The agreement says nothing about how much the dues are. The dues are set by the management board annually.
2. The agreement says nothing about shares or half shares.
3. The agreement says nothing about the allocation of beamtime.
4. You can opt out by giving 1 year’s notice.

NU is waiting to hear who is on board and who from your institution is signing the document.

Preparation for the Eiger 9M

There are a few things that we need to do to support this new detector:

1. Make the mounting brackets.
2. Make a protective device to keep people from messing with it and to keep the direct beam from hitting it.
3. Upgrade portions of the network to 10G
4. Increase the speed and capacity of at least part of the data storage system.

The hardware additions are minimal ($~5K). The network and computing infrastructure costs are likely to run ~$100K based on the current prices for network switches, disk drives, and servers. We'll be working with the manufacturer to be sure what we supply is well suited to the detector.

Budget FY 16 (Flat Option)

Salaries			
Base	882,579		100% KB, JSB, MB, JVO, EK, DS, SA, ZW
Fringe	241,826		95% NB
Total		1,124,405	
Items w/o F&A			
Capital Eq.	0		
Rent	120,161		
Telcom	8,000		
Total		128,161	
Items w/F&A			
Lab Supplies	24,850		
Office Supplies	2,500		
Lab Services	27,000		
Insurance	18,000		
Travel	0		
Total		72,350	
F&A		374,584	
Total		1,699,500	
Per Share		309,000	⇐ NU, UM, MSU, VARI
Per 0.5 Share		154,500	⇐ UI, VU, UW

Vote 1:
Approval of amounts shown.

Vote 2:
The capital equipment fund (~$800K) will be used for maintenance whenever it is needed. This fund will also be used to implement network and computing upgrades to support the Eiger detector. These upgrades are estimated to be about $100K.

The board will be kept informed about unusually large expenses.

Budget FY 16 (3% Option)

Salaries			
Base	882,579		100% KB, JSB, MB, JVO, EK, DS, SA, ZW
Fringe	241,826		95% NB
Total		1,124,405	
Items w/o F&A			
Capital Eq.	29,780		
Rent	120,161		
Telcom	8,000		
Total		157,941	
Items w/F&A			
Lab Supplies	36,000		
Office Supplies	2,500		
Lab Services	27,000		
Insurance	18,000		
Travel	5,000		
Total		88,500	
F&A		379,639	
Total		1,750,485	
Per Share		318,270	⇐ NU, UM, MSU, VARI
Per 0.5 Share		159,135	⇐ UI, VU, UW

Vote 1:
Approval of amounts shown.

Vote 2:
We expect to have more capital expenses than has been allocated here due to network and computing infrastructure improvements (~$100K) as well as our normal maintenance activities. The Capital Equipment fund (~$800K) will be tapped when the Capital Eq. line here is depleted.

The board will be kept informed about unusually large expenses.

Budget FY 16 (325 Option)

Salaries			
Base	882,579		100% KB, JSB, MB, JVO, EK, DS, SA, ZW
Fringe	241,826		95% NB
Total		1,124,405	
Items w/o F&A			
Capital Eq.	41,848		
Rent	120,161		
Telcom	8,000		
Total		170,009	
Items w/F&A			
Lab Supplies	45,000		
Office Supplies	2,500		
Lab Services	32,000		
Insurance	18,000		
Travel	10,000		
Total		107,500	
F&A		385,586	
Total		1,787,500	
Per Share		325,000	⇐ NU, UM, MSU, VARI
Per 0.5 Share		159,135	⇐ UI, VU, UW

Vote 1:
Approval of amounts shown.

Vote 2:
We expect to have more capital expenses than has been allocated here due to network and computing infrastructure improvements (~$100K) as well as our normal maintenance activities. The Capital Equipment fund (~$800K) will be tapped when the Capital Eq. line here is depleted.

The board will be kept informed about unusually large expenses.